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# Allogeneic cetuximab-armed gamma delta T cells using antibody-cell conjugation technology for the treatment of EGFR-expressing solid tumors

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## ABSTRACT

**Background** Targeting epidermal growth factor receptor (EGFR) has become a strategic approach in cancer therapy, using various modalities including chimeric antigen receptor (CAR)-αβT cell therapies. Despite significant advancements in autologous CAR-αβT cell therapies in B-cell lymphoma, current cell therapies face challenges such as potential risks associated with genetic engineering, waiting time and high costs of autologous CAR-αβT cell therapies. Innovations in click chemistry and bioorthogonal chemistry have enabled the development of antibody-cell conjugation (ACC) technology, which links cancer-targeting antibodies to immune cells without genetic modifications, potentially providing a safer profile.

**Methods** In this study, we introduce ACE2016, an innovative allogeneic cell therapy targeting EGFR. ACE2016 is generated by ACC technology to conjugate donor-derived γδ2 T cells with the EGFR-specific antibody cetuximab.

**Results** Our preclinical studies demonstrate that ACE2016 exhibits superior cytotoxicity against various EGFR-expressing cancer cell lines and minimal cytotoxic effects on normal cells. Mechanistic studies revealed that ACE2016 enhances cytotoxicity through increased capacity towards EGFR-expressing cancer cells, enhanced levels of cytotoxic cytokines and recruitment of peripheral cytotoxic cells, reflecting significant tumor suppression and prolonged survival in ACE2016-treated groups without causing treatment-related toxicity in vivo.

**Conclusions** These findings support the clinical potential of ACE2016 as an off-the-shelf γδ2 T-cell therapy for EGFR-expressing cancers, offering a combination of specificity, scalability, and safety in the development of solid tumor therapy.

### WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The success of autologous chimeric antigen receptor (CAR)-αβT cell therapies in B-cell lymphoma brings a glimmer of hope to patients with cancer; however, current cell therapies still face challenges such as potential risks associated with genetic engineering, long waiting time and high costs of autologous CAR-αβT cell therapies.

### WHAT THIS STUDY ADDS

⇒ By applying antibody-cell conjugation technology to allogeneic donor-derived γδ2 T cells, a novel off-the-shelf epidermal growth factor receptor (EGFR)-targeted γδ2 T cells, ACE2016, was developed without a need for genetic engineering. This study illustrates the superior efficacy of ACE2016 against solid tumors with various EGFR expression and the activation mechanism of ACE2016 on encountering target cells.

### HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The clinical potential of ACE2016 is currently investigated in a phase I trial. ACE2016 may provide a promising off-the-shelf, affordable and safe cell therapy for patients with EGFR-expressing cancers.

## INTRODUCTION

Epidermal growth factor receptor (EGFR) is a receptor tyrosine kinase within the ErbB family, playing a pivotal role in the pathogenesis of various malignancies, including non-small cell lung cancer, triple-negative breast cancer, head and neck squamous cell carcinoma, colorectal cancer, and glioblastoma. Overexpression or aberrant activation of EGFR enhances tumorigenesis by promoting cellular proliferation, survival, invasion, and metastasis.[1] Consequently, targeting EGFR has become a strategic approach in targeted cancer therapy, using modalities such as tyrosine kinase inhibitors,[2 3] monoclonal antibodies,[4 5] antibody-drug conjugates,[5–7] bispecific immune cell engagers,[8] and chimeric antigen receptor (CAR)-modified immune cells.[9–12]

CAR-αβT cell therapy, involving the genetic engineering of αβ T cells to target EGFR, has demonstrated promising safety and clinical efficacy in phase I trials.[9–11] Similarly, CAR-natural killer (NK) cells, engineered to target EGFR, offer advantages such as reduced risk of cytokine release syndrome and graft-versus-host disease (GvHD), and are currently in the preclinical stage, showing potential clinical efficacy.[12] Despite these advancements, significant challenges remain. Autologous CAR-αβT cell therapies require T cells from patients, which can be logistically challenging, time-consuming, and costly. Additionally, patients with compromised immune systems may not produce sufficient functional T cells. Allogeneic cell therapies, derived from healthy donors, offer a promising solution by being pre-manufactured, stored, and readily available for 'off-the-shelf' use.[13] This reduces treatment delays and costs, and allows for multiple dosing and combination therapies.[13] However, the involvement of genetic engineering in autologous CAR-αβT cell therapies introduces the potential risk of secondary cancer development.[14] A technology without the need for genetic engineering could provide a better safety profile for patients.

Innovations in click chemistry and bioorthogonal chemistry, recognized with the Nobel Prize in 2022, have revolutionized biomedical research by allowing the selective modification of biomolecules and living cells without interfering with biological processes.[15] Hsiao *et al* applied bioorthogonal chemistry to develop antibody-cell conjugation (ACC) technology to link cancer-targeting antibodies to immune cells including cytokine-induced killer cells, NK cells, and γδT cells.[16–19] The successful application of ACC technology in cancer treatment has been demonstrated by the specific recognition of cancer antigens and the superior anticancer potency of antibody-armed NK cells, leading to clinical trials for HER2-expressing tumors (NCT04319757) and CD20-targeting γδ2 T cells to B lymphoma (NCT05653271).[16–19]

In this study, we introduce an innovative allogeneic cell therapy targeting EGFR developed by the application of ACC technology. We generated cryopreserved cetuximab-linked γδ2 T cells, ACE2016, and evaluated their potency against EGFR-expressing cancer cells. This study elucidates ACE2016 as a potent and off-the-shelf cell therapy against EGFR-expressing cancers, addressing current gaps in EGFR-targeted treatments. This novel therapy combines the specificity of antibody targeting with the scalability and practicality of allogeneic cell products, supporting the potential clinical application of ACE2016.

## MATERIALS AND METHODS

### Antibodies, cell lines, and mice

All antibodies used in this study were obtained from BioLegend, except for anti-F(ab')2 antibody, which was sourced from Jackson Immuno Research Labs. Human lung cancer cell lines (HCC827 and A549), breast cancer cell line (MDA-MB-231), liver cancer cell line (HepG2), and human normal keratinocyte (HEKa), fibroblast (HLF and HUF), epithelial (HSAEC and RPTEC), and skeletal muscle (HSKMC) cells were obtained from JCRB or ATCC and cultured according to standard protocols. Peripheral blood mononuclear cells (PBMCs) from healthy donors were purchased from Charles River Laboratories. Female SCID-Beige (CB17.Cg-*Prkdc*$^{scid}$*Lyst*$^{bg\text{-}J}$/Crl, 6–10 weeks old) mice were also sourced from Charles River Laboratories and were housed under the regulation of the Institutional Animal Care and Use Committee (IACUC) of the contract research organizations.

### ACE2016 generation

γδ2 T cells from PBMCs (Charles River) of healthy donors were specifically activated by 1 µM of zoledronate and further expanded by 350 IU/mL of interleukin (IL)-2 for 16 days; to ensure the purity of γδ2 T cells, αβ T cells were depleted using T-cell receptor (TCR)α/β T Cell Depletion Kit (Miltenyi Biotec) according to the manufacturer's instructions (Patent WO2022/221506, 18). To prioritize patient safety, the health status of donors was evaluated through medical history review and adventitious virus testing; to minimize potential variability in potency, the anticancer activity of ACC-linked γδ2 T cells was assessed by in vitro cytotoxicity analysis against cancer cells to ensure the consistent quality of ACC products. Following a 16-day γδ2 T-cell expansion period, the conjugation of anti-EGFR antibody cetuximab to expanded γδ2 T cells was performed following previously described protocols (Patent WO2015/168656, 16, 19). In brief, NHS-DNA conjugates were generated from 5'-thiolated single-strand DNA (ssDNA) linker-1 and linker-2 with complementary sequences, and were then conjugated with cetuximab and γδ2 T cells, respectively. The linker-1-conjugated cetuximab and linker-2-conjugated γδ2 T cells were mixed for complementary DNA hybridization. After removing unbound linker-1-conjugated cetuximab, the cetuximab-linked γδ2 T cells were cryopreserved and stored in liquid nitrogen for subsequent analysis.

### Flow cytometry analysis

Cell suspensions derived from cells stained with fluorescent dye-conjugated antibodies were analyzed using the Attune NxT Flow Cytometer installed with Attune NxT software V.3.1.0, after washing with Dulbecco's phosphate-buffered saline (DPBS). propidium iodide (PI) staining was used to determine the viability of cells. γδ2 T cells were defined by gating CD3$^+$ and TCRδ2$^+$ population. The efficiency of cetuximab conjugation on γδ2 T cells was evaluated by staining with anti-F(ab')2 antibody (Jackson Immuno Research Labs).

To examine EGFR binding capacity, cells were incubated with 0.001, 0.01, 0.1, 1, 10 and 100 µg/mL of recombinant human EGFR-His protein (ACROBiosystems). After washing with DPBS, the cells were stained with FITC-conjugated anti-6X His tag antibody (Abcam). The stained cells were then washed by DPBS, and the


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EGFR-bound cell population was identified by flow cytometry analysis.

Human cancer cell lines (HCC827, A549, MDA-MB-231, and HepG2) and normal cells were stained with fluorescent dye-conjugated anti-EGFR antibody at corresponding concentrations. The EGFR expression of these stained cells, after washing with DPBS, was analyzed by flow cytometry.

## In vitro cytotoxicity

The CellTiter-Glo 2.0 Cell Viability Assay was performed according to the manufacturer's instructions. In brief, EGFR-positive HCC827, A549, and MDA-MB-231, along with EGFR-negative HepG2 cancer cell lines, were seeded in white opaque plates and incubated with the effector cells at effector (E) to target (T) ratios indicated in figure legends. The cultures were then mixed with the reaction substrate, and luminescence was detected by Synergy H1 Hybrid Multi-Mode Reader (BioTek). The absorbance of co-culture of effector (E) and target (T) cells, effector alone, and target alone was recorded. The percentage of cytotoxicity of each group was calculated using the equation $(1 - (ET - E) / T) \times 100$.

To evaluate the cytotoxicity of ACE2016 against PBMCs, a flow cytometry-based assay was employed. PBMCs were labeled with fluorescent CellTrace CSFE labeling dye (Invitrogen) and co-cultured with ACE2016 and γδ2 T cells at indicated effector to target ratios at 37°C for 4 hours. After co-culture, the cells were stained with PI, and the percentage of $PI^+$ cells within the carboxyfluorescein succinimidyl ester (CFSE)-labeled population was determined. The percentage of $PI^+CFSE^+$ cells was used to quantify the cytotoxicity exerted by ACE2016 and γδ2 T cells.

## Degranulation and cytokine production analysis

For the analysis of secreted interferon (IFN)-γ, tumor necrosis factor (TNF)-α, IL-8, and IL-10, a beads-based multiplex cytokine assay (BioLegend) was performed according to manufacturer's instruction. The cells were further prepared and analyzed for intracellular effector function markers CD107a, granzyme B and IFN-γ as previously described.[17] In brief, the co-cultured cells were harvested, fixed, and permeabilized before staining with fluorescent dye-conjugated antibodies against the respective markers. The stained cells were analyzed using the Attune NxT Flow Cytometer equipped with Attune NxT software V.3.1.0. The fold change in mean fluorescence intensity of intracellular CD107a, granzyme B, and IFN-γ in effector cells was calculated relative to γδ2 T cells in the absence of target cancer cells (baseline set as 1).

## Immune cell recruitment assay

Effector (ACE2016 and γδ2 T cells) and target (MDA-MB-231) cells were co-cultured at a ratio of 2:1 for 24 hours. The culture supernatant was collected as a chemoattractant in the outer chamber of the culture insert. PBMCs were seeded in the inner chamber of the culture insert. The cultures were incubated at 37°C for 24 hours. After incubation, the migrated cells, including T cells ($CD3^+$), $CD4^+$ T cells ($CD4^+CD3^+$), and $CD8^+$ T cells ($CD8^+CD3^+$), were detected using flow cytometry analysis.

## Animal studies

To evaluate the in vivo efficacy of ACE2016, $5\times10^6$ HCC827 cells were subcutaneously and $5\times10^6$ MDA-MB-231 cells were orthotopically implanted into female SCID-Beige mice to evaluate the in vivo efficacy of ACE2016 in the xenograft lung and breast cancer models, respectively. All animal experiments were done with approval from the IACUC at OneNess Biotech (IACUC No. 220511012, 230 211 005 and 230811029) and Pharmacology Discovery Services (IACUC No. ODS112-01053887). When the tumor volume reached 50–100 $mm^3$, the tumor-bearing mice were randomly assigned to groups. ACE2016 ($1\times10^7$), γδ2 T cells ($1\times10^7$), and vehicle were administered intravenously through the tail vein two times a week for 2 weeks. The tumor volume was calculated using the formula: length×width×width×0.52 and monitored along with the survival of the mice. All mice were sacrificed 56–95 days post the first dose or on reaching the humane endpoints. Immunohistochemistry of tumor-infiltrated ACE2016 and γδ2 T cells was performed and analyzed as described previously.[20] In brief, formalin-fixed, paraffin-embedded (FFPE) organ blocks were sliced and stained with anti-TCRδ antibody (Santa Cruz, clone H-41) at 1:50 dilution followed by detection with mouse/rabbit PolyDetector Plus DAB HRP kit (Bio SB). Slides were counterstained with hematoxylin (Leica).

To analyze the safety of ACE2016 in vivo, ACE2016 ($1\times10^7$), γδ2 T cells ($1\times10^7$), and vehicle were intravenously administered to female SCID-Beige mice according to the regimen used in the efficacy study. Treated mice were monitored for clinical observations throughout the study and were sacrificed either 1 day after the last dose or after a 14-day recovery period. Necropsies were performed, and blood samples from each mouse were collected for subsequent hematology (n=5) and serum chemistry (n=5) analyses.

## Statistical analysis

Data from the cytotoxicity assay and flow cytometry were analyzed using an unpaired Student's t-test. Tumor luminescence signals between each treatment in animal studies were analyzed using two-way analysis of variance.

# RESULTS

## Cetuximab conjugation on γδ2 T cells using ACC technology

To confer EGFR-targeting specificity to γδ2 T cells, ACC technology was used to conjugate γδ2 T cells from PBMCs of healthy donors with anti-human EGFR specific antibody cetuximab (figure 1a). ssDNA linker-1 conjugated cetuximab and ssDNA linker-2 conjugated γδ2 T cells were linked to form cetuximab-armed γδ2 T cells, termed ACE2016.

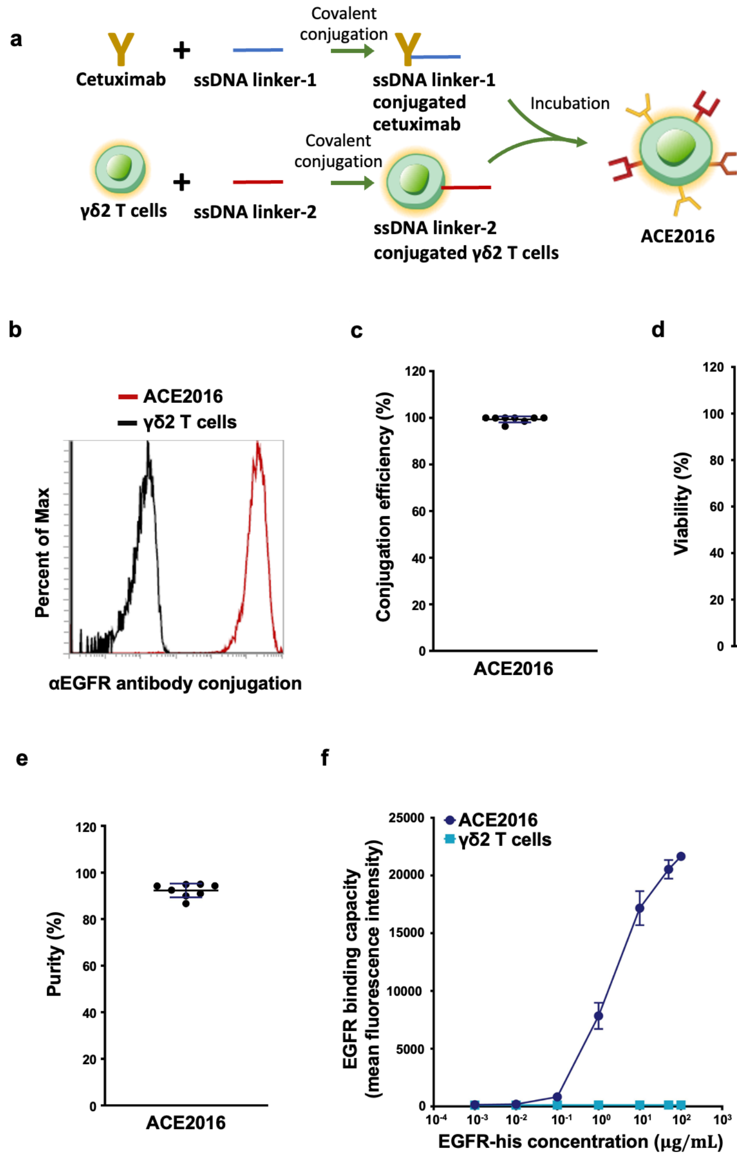


**Figure 1** Cetuximab-conjugated γδ2 T cells exhibit EGFR binding capacity. (a) The flow chart of ACE2016 preparation is shown. Donor peripheral blood mononuclear cell-derived activated γδ2 T cells were enriched and conjugated with anti-EGFR antibody cetuximab, followed by cryopreservation to generate ACE2016. (b) A representative histogram of cetuximab conjugation is illustrated. Un-conjugated γδ2 T cells and ACE2016 were stained with anti-F(ab')2 antibody to determine the cetuximab conjugation efficiency. The dark line represents the negative staining of unconjugated γδ2 T cells, while the red line indicates efficient cetuximab conjugation on ACE2016. (c, d, e) Efficient cetuximab conjugation with γδ2 T cells and (d) consistently decent viability were observed in independent experiments (n=8, from five donors). (f) EGFR binding capacity of ACE2016 and γδ2 T cells was determined by flow cytometry analysis. Cells were incubated with 0.001, 0.01, 0.1, 1, 10 and 100 µg/mL of human EGFR-His recombinant protein, and the EGFR-bound cell population was identified by staining with FITC-conjugated anti-6X His tag antibody. Each condition was performed in triplicate, and data are presented as mean value±SD. EGFR, epidermal growth factor receptor; ssDNA, single-strand DNA.

Cetuximab was consistently conjugated to γδ2 T cells with approximately 100% efficiency (figure 1b,c), without affecting the viability and purity of ACE2016 (figure 1d,e) across eight independent preparations from five donors. To characterize the specificity and purity of ACE2016 binding to EGFR, the binding capacity of ACE2016 to human EGFR-His recombinant protein at designated concentrations was analyzed by flow cytometry. As shown in figure 1f, ACE2016 exhibited significantly increased binding to human EGFR recombinant protein compared with unconjugated γδ2 T cells.

## Enhanced specific cytotoxicity of ACE2016 against EGFR-expressing cancer cells

To examine the potency of ACE2016 against EGFR-expressing cancer cells, ACE2016 was co-incubated with HCC827 (high EGFR expression), A549 (medium EGFR expression), MDA-MB-231 (low EGFR expression), and HepG2 (minimal to no EGFR expression) (figure 2a).

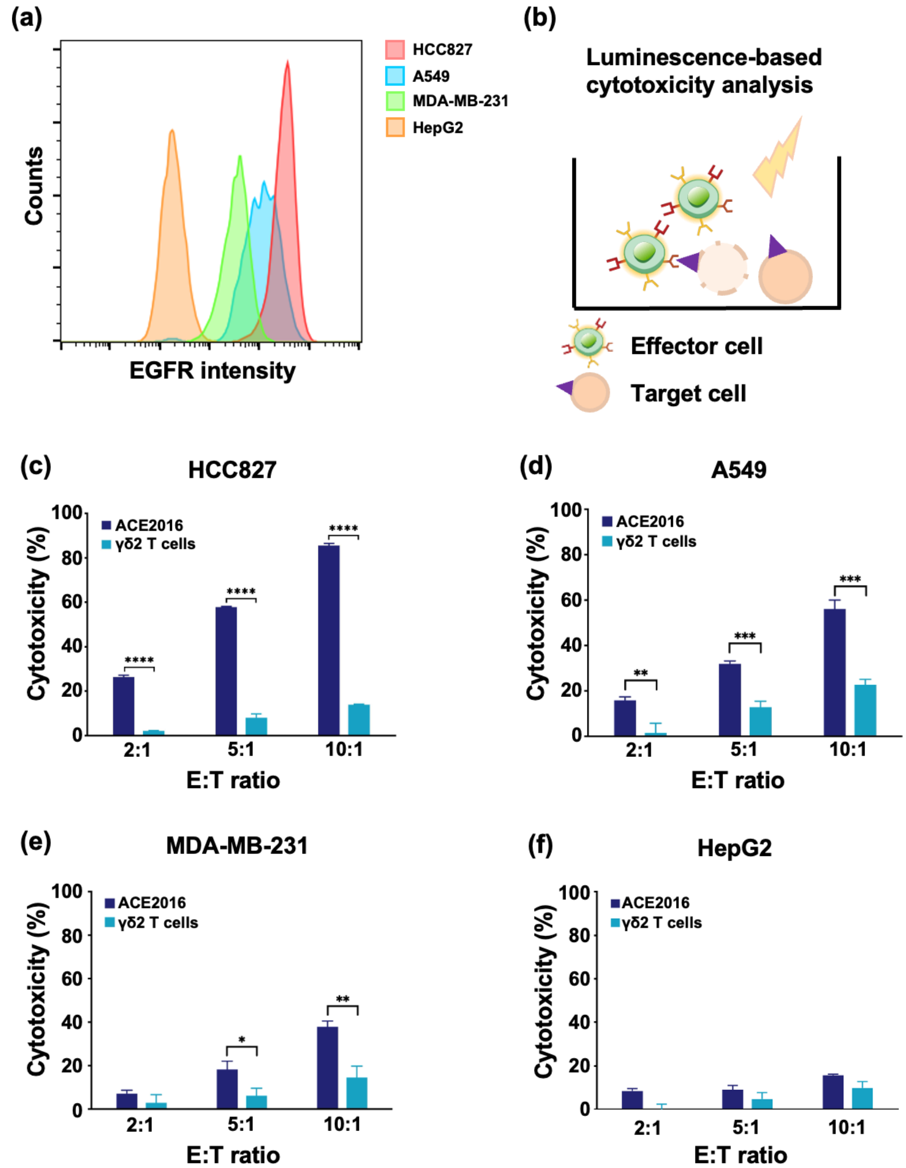


**Figure 2** Cetuximab-conjugated γδ2 T cells elicit superior cytotoxicity against EGFR-expressing cancer cells. (a) Representative histograms of EGFR expression in HCC827, A549, MDA-MB-231, and HepG2 cells are shown. (b) The design of the cell cytotoxicity assay design is illustrated. ACE2016 and γδ2 T cells were co-incubated with (c) HCC827, (d) A549, (e) MDA-MB-231, and (f) HepG2 cells at effector to target (E:T) ratio of 2:1, 5:1 and 10:1, and analyzed by CellTiter-Glo luminescent cell viability assay after 4 hours of co-incubation. All experiments were performed in triplicate. Data were presented as mean value±SD. Statistical analysis was performed using t-test. *p<0.05; **p<0.01; ***p<0.001; ****p<0.0001. EGFR, epidermal growth factor receptor.

Cytotoxicity was measured using the CellTiter-Glo luminescent cell viability assay (figure 2b). The results revealed that ACE2016 elicited significantly enhanced cytotoxicity against HCC827, A549, and MDA-MB-231 cells compared with unconjugated γδ2 T cells at effector (E) to target (T) ratios of 2:1, 5:1 and 10:1 (figure 2c–2e). ACE2016 showed no significant cytotoxicity against HepG2 cells (figure 2f). These results suggest the specific potency of ACE2016 against cancer cells with low to high EGFR levels.

To assess the on-target off-tumor cytotoxicity of ACE2016 to human normal EGFR-expressing cells, the EGFR levels of human normal keratinocytes (HEKa), fibroblasts (HLF and HUF), epithelial cells (HSAEC and RPTEC), and skeletal muscle (HSKMC) cells were first determined (figure 3a). These normal EGFR-expressing cells and HCC827 cells were incubated with ACE2016 and unconjugated γδ2 T cells. The results demonstrated that ACE2016 exhibited enhanced cytotoxicity against HCC827 cells, low cytotoxicity against HEKa and HLF normal cells, and no cytotoxicity against the remaining normal cells with low to no EGFR expression (HSAEC, HSKMC and PRTEC, figure 3b). When the PBMCs from healthy donors were co-cultured with ACE2016 or unconjugated γδ2 T cells, no significant off-target cytotoxicity

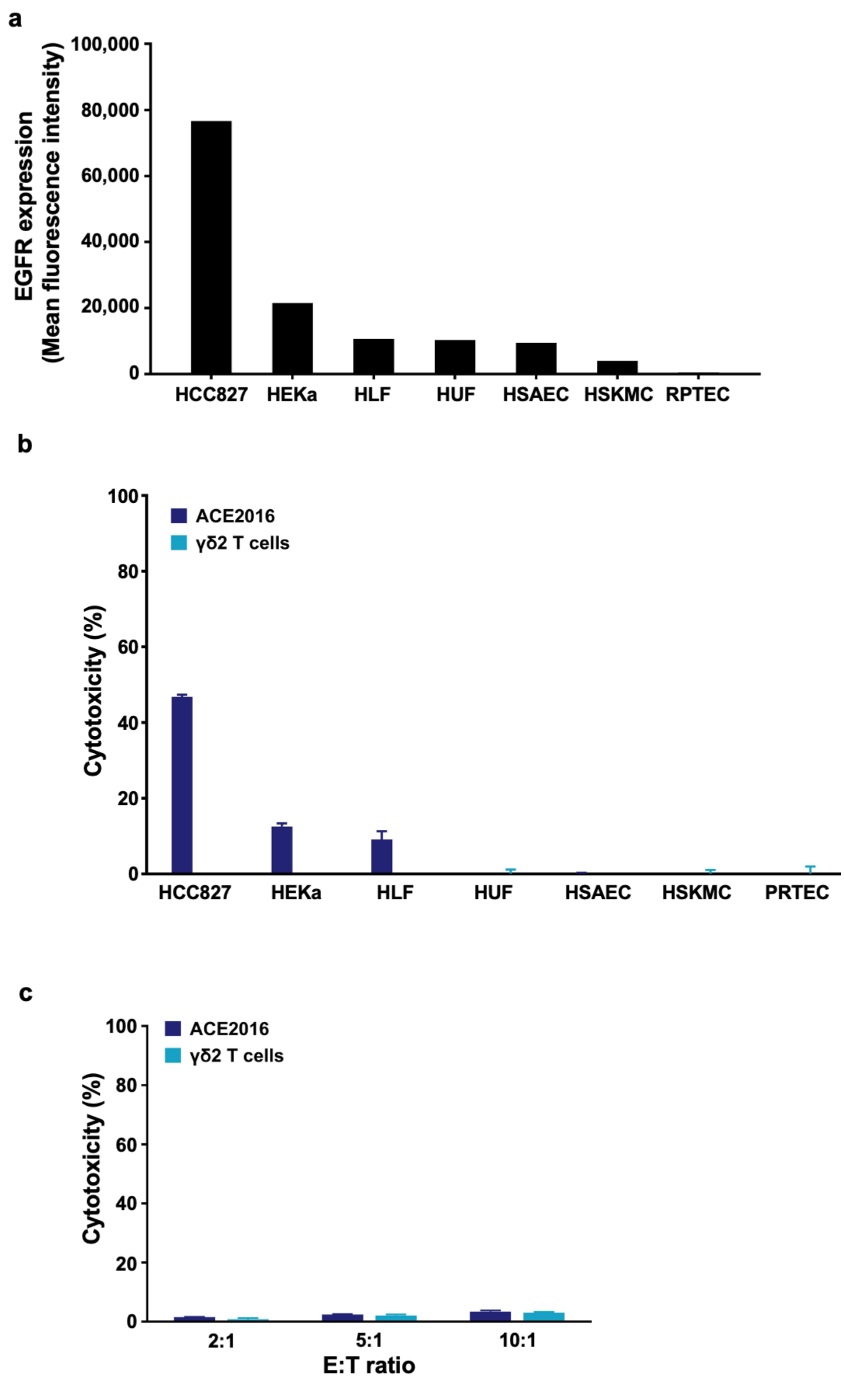


**Figure 3** ACE2016 exhibits minimal on-target, off-tumor cytotoxicity to EGFR-expressing primary normal cells. (a) EGFR expression in human normal keratinocyte (HEKa), fibroblast (HLF and HUF), epithelial (HSAEC and RPTEC), and skeletal muscle (HSKMC) cells was examined. Cells were incubated with fluorescent dye-conjugated anti-EGFR antibody at 2 µg/mL and detected by flow cytometry. The EGFR staining of the HCC827 cancer cell line was used as a reference. (b) The cytotoxicity of ACE2016 and γδ2 T cells against HCC827 and human primary normal cells was assessed. Target cells are sorted from left to right from high to low EGFR expression. Effector to target ratio at 5:1 was co-cultured and analyzed by CellTiter-Glo luminescent cell viability assay after 48 hours of co-incubation. (c) ACE2016 and γδ2 T cells were co-incubated with fluorescent dye-labeled peripheral blood mononuclear cells at effector to target (E:T) ratios of 2:1, 5:1, and 10:1, and analyzed by flow cytometry-based cell viability assay after 4 hours of co-incubation. All experiments were performed in triplicate. Data are presented as mean value±SD. EGFR, epidermal growth factor receptor.

elicited by ACE2016 was observed (figure 3c). These results preliminarily revealed tolerable cytotoxicity against EGFR-expressing normal cells by ACE2016.

### Activation of ACE2016 when encountering EGFR-expressing cancer cells

To delineate the mechanisms underlying ACE2016's cytotoxicity against EGFR-expressing cancer cells, we applied a multiplex cytokine assay and flow cytometry to investigate the secreted and intracellular cytokines from ACE2016, respectively. As shown in figure 4a–4c, there was significantly enhanced secretion of IFN-γ, TNF-α, and IL-8 in the co-culture of ACE2016 with MDA-MB-231 and HCC827 cells compared with unconjugated γδ2 T cells. CD107a, granzyme B, and IFN-γ are known effector function markers for the cytolytic activity of γδ T cells and other cytotoxic immune cells when encountering cancer cells.[21–23] Intracellular staining demonstrated that production of CD107a, granzyme B and IFN-γ was significantly elevated in ACE2016 co-cultured with MDA-MB-231 cells (figure 4d–4f), suggesting that ACE2016 is activated and secretes cytotoxic cytokines on encountering EGFR-expressing cancer cells. Furthermore, we investigated the migratory capacity and capability of recruiting other cytotoxic cells. The supernatant from ACE2016 co-cultured with MDA-MB-231 cells showed the ability to recruit T cells including $CD4^+$ and $CD8^+$ T cells (figure 4g). Taken together, these results suggested that ACE2016 recruits cytotoxic immune cells by increased cytokine secretion when encounters EGFR-expressing cancer cells.

### In vivo potency of ACE2016 against EGFR-expressing cancer cells

To examine the antitumor activity of ACE2016 against EGFR-expressing cancer cells in vivo, the potency of ACE2016 was evaluated in one subcutaneous xenograft model of lung cancer cell line HCC827 and one orthotopic xenograft model of triple-negative breast cancer cell line MDA-MB-231. ACE2016 and γδ2 T cells were infused two times a week for 2 weeks into SCID-Beige mice bearing HCC827 or MDA-MB-231 (figure 5a). The tumor volume, body weight, clinical signs, and survival status were monitored. ACE2016 exhibited strong tumor suppression to the xenograft models HCC827 and MDA-MB-231 compared with vehicle and γδ2 T cells (figure 5b), and the reproducible results were observed in three independent studies for HCC827 and MDA-MB-231 xenograft models (figure 5c). Bioluminescence results further support substantial tumor growth suppression of HCC827 tumor by ACE2016 (online supplemental figure 1), and ACE2016 displayed the capacity of tumor infiltration as shown by immunohistochemistry (IHC) analysis (figure 5d). Importantly, the strong antitumor activity was reflected in the prolonged survival of ACE2016-treated mice (figure 5e), and ACE2016 treatment did not cause changes in body weight during the studies (figure 5f). These results demonstrated the in vivo antitumor capacity of ACE2016 in different EGFR-expressing cancer models.

In addition to assessing antitumor activity, we further conducted toxicology evaluations, including clinical observation, macroscopic examination of primary organs (brain, heart, lung, spleen, liver, and kidneys), serum chemistry, and hematology analyses. These evaluations were performed during both an acute phase and a 14-day recovery phase toxicology study (online supplemental figure 2). Following two times a week intravenous injections of ACE2016 for 2 weeks, the results revealed that multiple administrations of ACE2016 had no impact on the weight of body and organs (lung, heart, liver, spleen, and kidney) throughout the study. Additionally, hematology analysis showed no observable changes in the percentages of lymphocytes, monocytes, red blood cells, platelets, basophils, eosinophils, and neutrophils in the ACE2016-treated group compared with vehicle-treated and control γδ2 T cells-treated groups. These observations further supported the safety of ACE2016 administration.

## DISCUSSION

Bioorthogonal chemistry enables rapid, high-yielding, and selective chemical reactions in biological environments with minimal interaction with endogenous functional groups.[24] Using copper-free click chemistry, this approach has revolutionized stable cell labeling without altering intrinsic characteristics, which is critical for in vivo cell tracking and targeted drug delivery in cancer therapy.[25 26] Our study highlights the transformative potential of bioorthogonal chemistry in immunotherapy, particularly through ACC technology developed by Hsiao *et al* (Patent WO2015168656). This technology combines rapid bio-conjugation reactions with the specificity and safety of DNA duplexes, offering a robust method for cell modification. It demonstrates compatibility with diverse antibodies and immune cell types, suggesting wide applications in targeted immune cell therapies.[16–19] In this study, multiple batches of ACE2016 from different donors were reproducibly generated, and the robust conjugation efficiency, viability and purity were shown in figure 1. Compared with conventional autologous CAR-αβT therapy, ACC products using a bioorthogonal chemistry-based approach require no genetic engineering, which significantly reduces the complexity, cost and time of manufacturing and secondary cancer risk. Sustainable source of allogeneic γδ2 T cells further supports the off-the-shelf application and reduces the waiting time for patients. In addition to ACE2016 (NCT06415487), one NK-based (ACE1702, NCT04319757) and one γδ2 T cells-based (ACE1831, NCT05653271) cell therapeutics based on a bioorthogonal chemistry-based approach have entered phase I trials. Tumor heterogeneity might be a hurdle to the clinical response for EGFR-targeting cell therapies including ACE2016. It has been reported that dual targeting and combination with other therapeutics would improve the clinical response of cell therapy.[27 28] Applying these two types of strategies would help ACE2016 to overcome tumor heterogeneity and acquired EGFR resistance and have a better clinical response. The feasibility of conjugating two cancer-targeting antibodies on γδ2 T cells using ACC technology is worthy of further investigation.

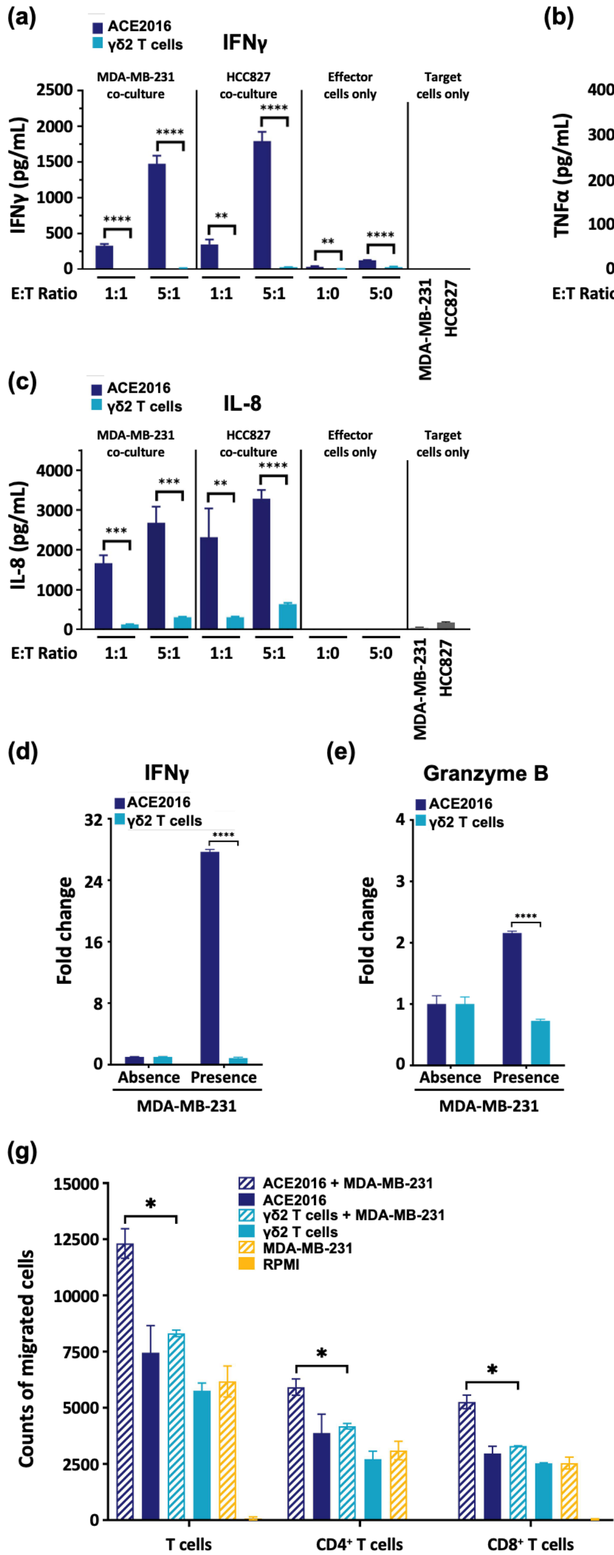


**Figure 4** ACE2016 is activated when encountering EGFR-expressing cancer cells. (a–d) ACE2016 and γδ2 T cells were co-incubated with cancer cell lines MDA-MB-231 and HCC827 at effector to target (E:T) ratios of 1:1 and 5:1 for 4 hours. The supernatant was harvested for detection of (a) IFN-γ, (b) TNF-α, and (c) IL-8 and IL-10 using multiplex cytokine analysis. (d-f) ACE2016 and γδ2 T cells were cultured in the absence and presence of MDA-MB-231 cells at an E:T ratio of 2:1 for 3 hours of co-incubation. The expression of intracellular (d) IFN-γ, (e) granzyme B, and (f) CD107a was analyzed by flow cytometry. (g) Conditional medium of co-culture of ACE2016 with target cells enhances the recruitment of cytotoxic immune cells from donor peripheral blood mononuclear cells. The conditioned medium derived from co-culture of effect to target ratio at 2:1 for 24 hours was used as a chemoattractant. All experiments were performed in triplicate. Data are presented as mean value±SD. Statistical analysis was performed using t-test. *p<0.05; **p<0.01; ***p<0.01; ****p<0.0001. IFN, interferon; IL, interleukin; TNF, tumor necrosis factor.

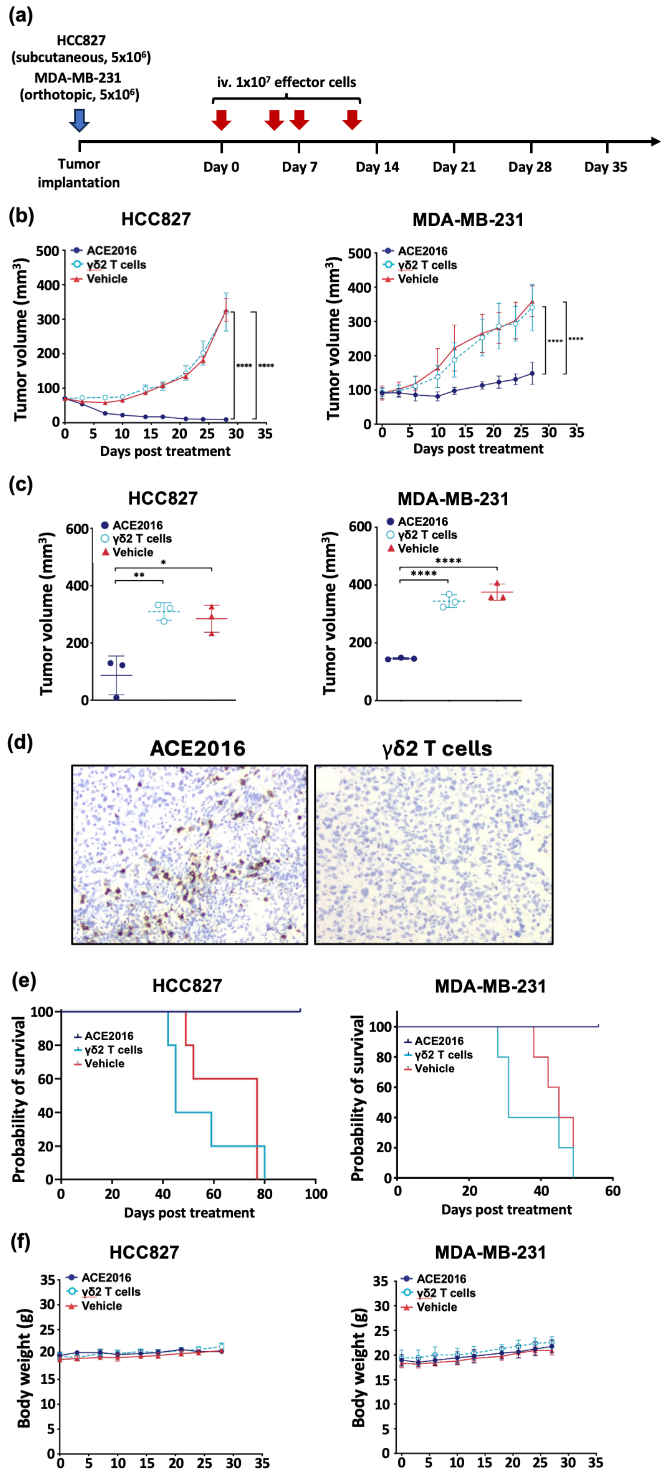


**Figure 5** ACE2016 displays superior in vivo potency against EGFR-expressing cancer cells. Four doses of intravenously delivered ACE2016 effectively suppressed tumor growth. Tumor-bearing SCID-Beige mice were treated with ACE2016, γδ2 T cells, and vehicle two times a week for 2 weeks. (a) Dosing regimen is illustrated. (b) Tumor volume in HCC827-bearing and MDA-MB-231-bearing SCID-Beige mice was monitored and presented as mean values±SD (n=5 per group). The difference in mean tumor burden between groups was examined by a two-way analysis of variance test. *p<0.05; **p<0.01; ****p<0.0001. (c) Three independent studies using HCC827-bearing and MDA-MB-231-bearing mouse models were performed. Statistical analysis was performed using t-test. *p<0.05; **p<0.01; ****p<0.0001. (d) Tumor infiltration of effector cells was examined by immunohistochemistry. Tumor biopsy samples were collected on day 38. FFPE organ blocks were sliced and stained with anti-T-cell receptor-δ antibody at 1:50 dilution followed by detection with mouse/rabbit PolyDetector Plus DAB HRP kit (Bio SB). Slides were counterstained with hematoxylin. (e) The survival of ACE2016-treated, γδ2 T cells-treated and vehicle-treated groups in HCC827-bearing and MDA-MB-231-bearing SCID-Beige mouse model was monitored. The survival rate of mice with different treatments was analyzed by the Kaplan-Meier method. (f) The body weight of HCC827 and MDA-MB-231-bearing mice was recorded and presented as mean values±SD. EGFR, epidermal growth factor receptor; FFPE, formalin-fixed, paraffin-embedded; iv, intravenous.

γδ T cells offer excellent chemotaxis, selective cell recruitment, and conservative monomorphism.[29–31] Activated γδ T cells provide multitarget recognition, potent cytotoxicity, and no major histocompatibility complex restriction, making them ideal for allogeneic adoptive immunotherapy.[32–34] Human γδ T-cell subsets exhibit tissue-specific tropism: γδ1-positive cells are enriched in mucosal tissues, while γδ2 T cells are enriched in the blood.[32] In this study, we successfully applied ACC technology to conjugate cetuximab with γδ2 T cells expanded from healthy donors, creating the EGFR-targeting ACE2016 product (figure 1). ACE2016 exhibited superior binding specificity to EGFR (figure 1f), significantly enhanced cytotoxicity against various EGFR-expressing cancer cells (figure 2c–2e) through upregulation of cytotoxic cytokines (figure 4a–4f). Without causing immunogenic potential when co-incubated with PBMCs (online supplemental figure 3), ACE2016 also elicited the least cytotoxic effects to PBMCs and EGFR-expressing normal cells (figure 3). ACE2016 significantly suppressed tumor growth, prolonged survival, and showed no adverse effects (figure 5 and online supplemental figure 2), highlighting its clinical potential for targeting EGFR-positive malignancies.

The kinase-activating mutations in EGFR or an overexpression of EGFR protein is observed in a broad spectrum of tumors, and therapies targeting EGFR and its downstream signaling cascades are under investigation.[35 36] Several tyrosine kinase inhibitors efficaciously elicit antitumor activity but eventually result in the development of acquired drug resistance in tumors. EGFR overexpression provides a chance for alternative therapeutics including EGFR-targeting antibodies, immunoconjugates, and cell therapies.[35 37 38] Over the past two decades, immunotherapy has remarkably improved clinical outcomes for various cancers. Among diverse strategies, immune effector cell retargeting has shown notable efficacy, with bispecific T-cell engagers (BiTEs) and CAR-αβT cells being the most prominent.[39] EGFR-directed antibody–drug conjugates (ADCs), such as AZD9592, MRG003, BL-B01D1, and M1231, are currently under clinical investigation. Among these, MRG003 and BL-B01D1 have been reported to have encouraging clinical efficacy and manageable safety profiles. These ADCs aim to maximize tumor cell killing while minimizing systemic toxicity.[6 7 40] EGFR-targeting bispecific cell engagers are also being actively investigated in preclinical and clinical trials to direct T or NK cells to attack EGFR-expressing tumor cells by simultaneously binding to EGFR and either CD3 or CD16. Current clinical trials including NCT05387265, NCT04903795, and NCT05099549 assess the safety and efficacy of these innovative treatments in various cancer types. Both ACE2016 and EGFR-targeting BiTE redirect immune cells to EGFR-expressing tumor cells through antibody-mediated antigen recognition and subsequent T-cell activation. However, BiTEs face challenges such as high manufacturing cost, short half-life, necessitating continuous intravenous administration, frequent dosing, upregulation of regulatory T cells, or anergic immune cells from patients.[41 42] ACE2016 addresses some of these challenges by reducing manufacturing cost and applying allogeneic γδ2 T cells from healthy donors. The tumor microenvironment (TME) poses additional obstacles for BiTEs, as endogenous T cells can become inactivated or anergic due to the presence of immune checkpoints like programmed cell death protein-1 (PD-1).[41] Moreover, immune cell infiltration is crucial for the success of immunotherapy. Cytotoxic γδ2 T cells have shown the ability to migrate from the bloodstream to tumor sites in both early and late cancer stages.[43] Zumwalde *et al* found that adoptively transferred γδ2 T cells bypass the immunosuppressive TME more effectively than $CD8^+$ and $CD4^+$ T cells due to lower PD-1 expression.[44] In an ovarian tumor-bearing mouse model, most tumor-infiltrating γδ T cells were of the γδ2 phenotype.[29] Aligning with these findings, ACE2016, which expands the γδ2 T-cell population from PBMCs of healthy donors, exhibited superior anticancer efficacy to EGFR-expressing cancers independent of wildtype or mutant status of EGFR and tumor tissue penetration compared with control γδ2 T cells (figure 5), underscoring its potential as an effective therapeutic option in cancer treatment.

CAR-αβT cell therapy is extending from autologous to allogeneic applications through genetic editing. A significant challenge in allogeneic settings is the risk of GvHD from donor T cells.[45] Gene editing, such as disrupting TCRαβ/CD3 genes, mitigates the risk but carries potential hazards like insertional mutagenesis, particularly when viral vectors are used.[46] Incomplete editing or off-target effects, even with advanced tools like CRISPR or TALEN, can still trigger GvHD. ACC technology presents a novel and promising alternative. By enabling the efficient conjugation of γδ2 T cells with specific antibodies like cetuximab, ACC technology circumvents the need for targeted cell therapies without genetic engineering. This approach enhances the cytotoxicity of γδ2 T cells against EGFR-expressing cells while maintaining a high safety profile by minimizing off-target effects (figure 3) and immunogenic potential (online supplemental figure 3). These advantages position ACC technology as a promising alternative for cancer immunotherapy.

The potential mechanism of action of ACE2016 against EGFR-expressing cancer cells is illustrated in figure 6. Previous study has demonstrated that BTN2A1 directly binds the $V\gamma9^+$ domain of the TCR, whereas a second ligand, potentially BTN3A1, binds the Vγ9Vδ2 TCR complex to activate γδ2 T cells.[47] In our recent study for ACE1831 (rituximab-linked γδ2 T cells), the mass spectrometry analysis identified

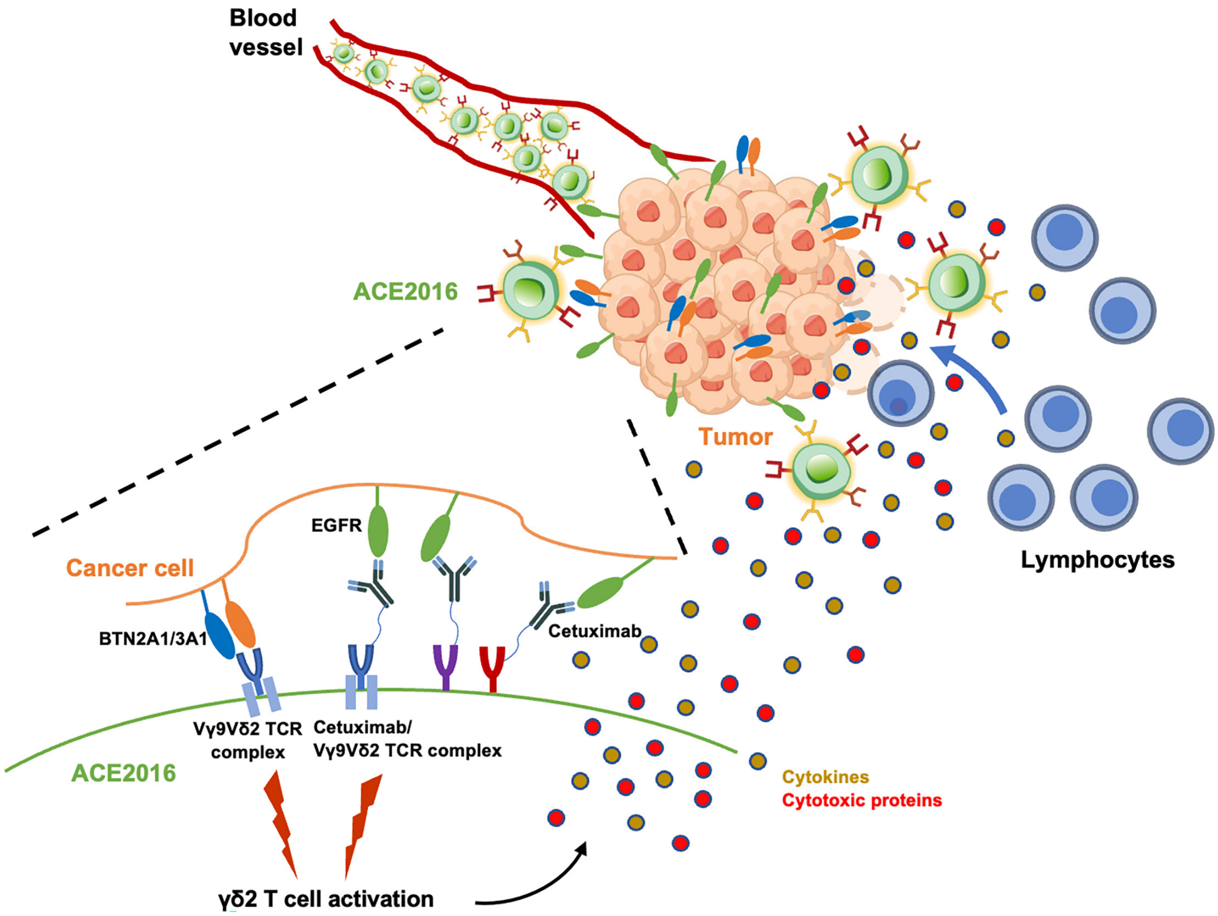


**Figure 6** Illustration of the mode of action of ACE2016 on encountering EGFR-expressing cancer cells. When ACE2016 encounters EGFR-expressing target cells, ACE2016 elicits cytotoxicity through upregulation of cytokines and cytotoxic proteins and degranulation. In the meantime, lymphocytes in the peripheral blood are recruited to the cytotoxic sites. EGFR, epidermal growth factor receptor; TCR, T-cell receptor.

multiple receptors, including TCR complex (TCRδ, CD3δ and CD3γ), to be linked with antibodies through ACC technology and might be involved in the functionality of ACC products.[18] Furthermore, the role of ACC-linked antibody/receptor complex and TCRγδ-BTN interaction in the potency of ACE1831 has been assessed.[18] The results showed that the blocking of TCRγδ-BTN interaction partially suppressed ACE1831-mediated cytotoxicity, suggesting that the ACC-linked antibody/receptor complex plays a critical role in ACE1831 recognizing and eliminating cancer cells, while TCRγδ-BTN interaction still makes a significant contribution to ACE1831 potency.[18] This study further showed anti-γδTCR blocking antibody partially decreased ACE2016 cytotoxicity to EGFR-expressing cancer cells (online supplemental figure 4), indicating both TCRγδ-BTN interaction and ACC-linked antibody/receptor complex play important roles in ACE2016 potency.

Additionally, increased levels of cytokines such as IFN-γ and TNF-α on co-culture with cancer cells underscore the activation of ACE2016 and their potent effector functions.[48] The markers of cytolytic activity, CD107a and granzyme B, were significantly elevated in ACE2016 when encountering cancer cells (figure 4a–4f), pointing towards an enhanced cytotoxic profile.[34] Cetuximab was detected on the surface of HCC827 co-cultured with ACE2016, implying potential antibody-dependent cellular cytotoxicity effect (online supplemental figure 5). The increased capacity of ACE2016 towards EGFR-expressing cancer cells and the recruitment of peripheral cytotoxic cells, including $CD4^+$ and $CD8^+$ T cells (figure 4g), suggest a potential bystander effect in the antitumor immune response.[49] The mechanisms described here are expected to confer ACE2016 with efficient and effective antitumor activity.

In summary, the ACC technology has been successfully applied to develop ACE2016, an EGFR-targeting γδ2 T-cell product, demonstrating superior in vitro and in vivo efficacy against EGFR-expressing cancer cells. The study elucidates the role of antibody/receptor complexes linked by ACC technology in activating γδ2 T cells and enhancing their cytotoxicity. As an allogeneic γδ2 T-cell therapy without genetic engineering, ACE2016 holds significant promise for treating patients with EGFR-expressing cancers, offering superior potency and a favorable safety profile.

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